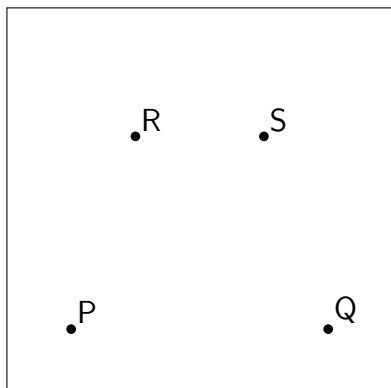
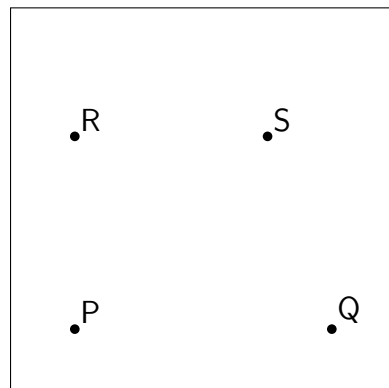


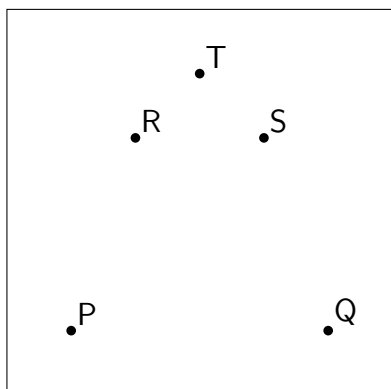
Problem 1: Trace each picture onto a separate square of thin paper. Make one straight fold that brings P onto Q and R onto S at the same time. Where T is shown, T must stay in its original place. Draw a crease that works, or explain why none can.



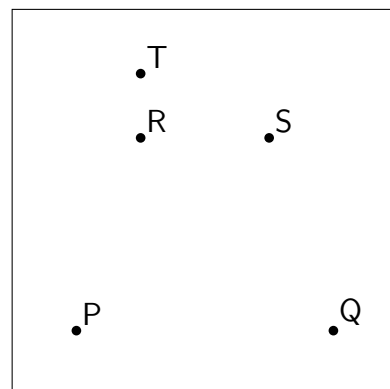
1



2

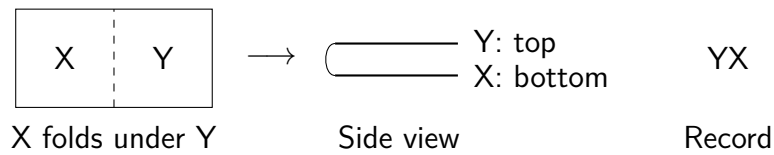


3



4

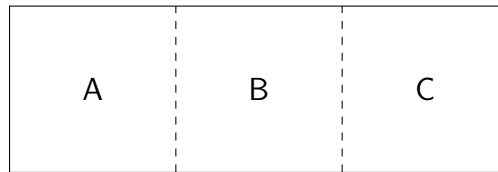
A strip's equal square panels stay connected. Fold only at the joins. Read a stack from top to bottom.



Keep the original front marked. “Toward the front” and “toward the back” refer to that original front at each join.

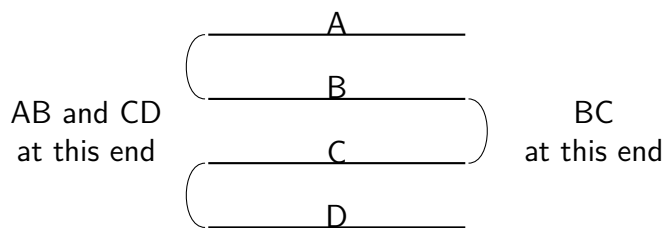
Read this three-panel strip with B’s original front facing up.

Problem 2: Fold a three-panel strip completely flat into a one-square stack. Find every top-to-bottom order of A, B, C. Can the same directions at both joins give different layer orders?



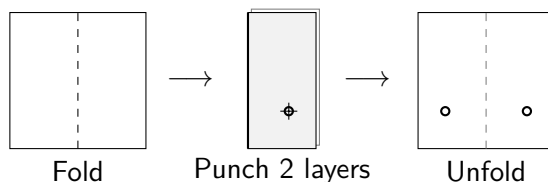
Layer order	Directions at the two joins

Problem 3: Arrange A, B, C, D as four thin layers in a flat stack. In a side view, joins AB and CD bend around the same end; join BC bends around the other end. Joining bends at the same end cannot cross. Find every layer order that obeys this rule.

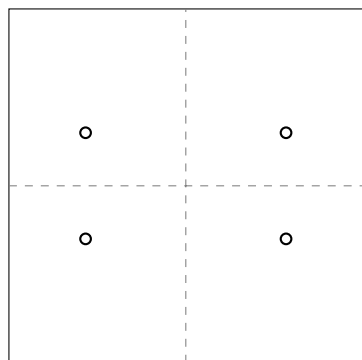
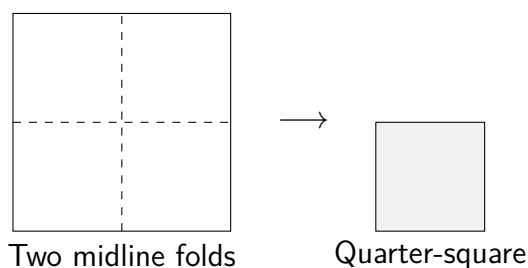


This is an ideal thin-paper side view. A noncrossing drawing does not supply a folding motion with real paper.

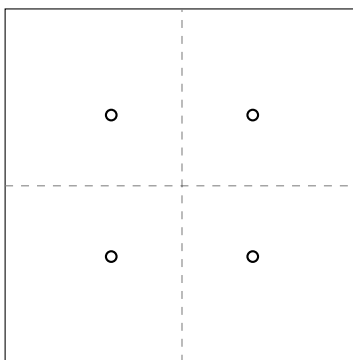
A punch goes through every layer. Unfold the paper to see its pattern.



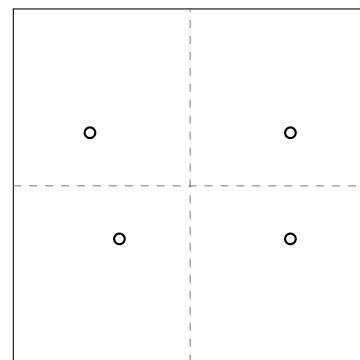
Problem 4: Fold a square on its vertical and horizontal midlines into a quarter-square. Keep the top-right quarter underneath. With an adult, make one punch through every layer, away from folds and edges. Which patterns below can come from that one punch? Mark the punch position on the folded quarter for each possible pattern.



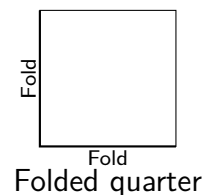
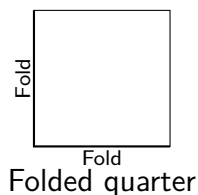
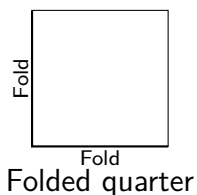
1



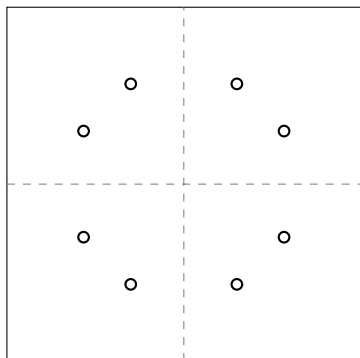
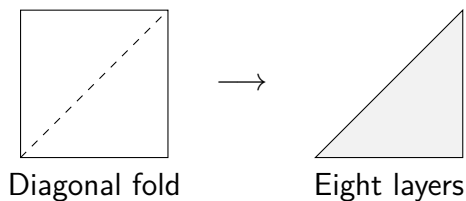
2



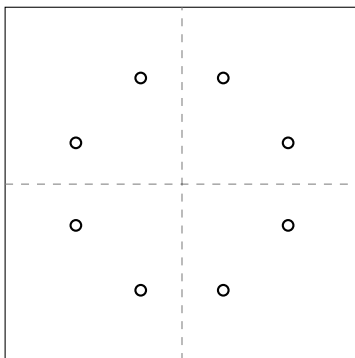
3



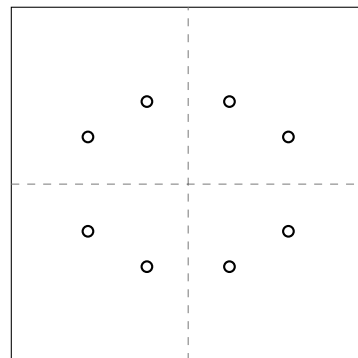
Problem 5: Fold the quarter-square from Problem 4 along its diagonal, making a triangle. Punch once through every layer, away from every fold and edge. Which patterns below can come from one punch? Mark a punch position for each possible pattern, or explain why it cannot work.



1



2



3

